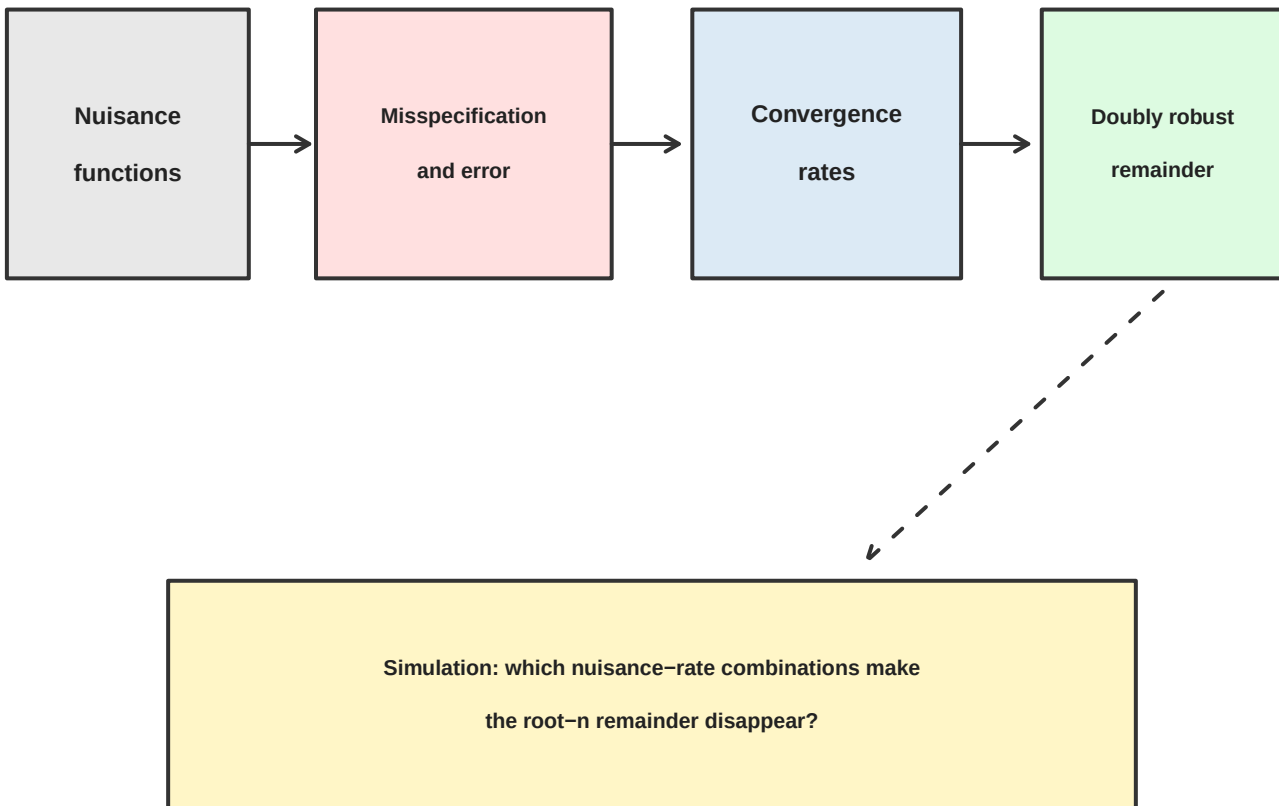
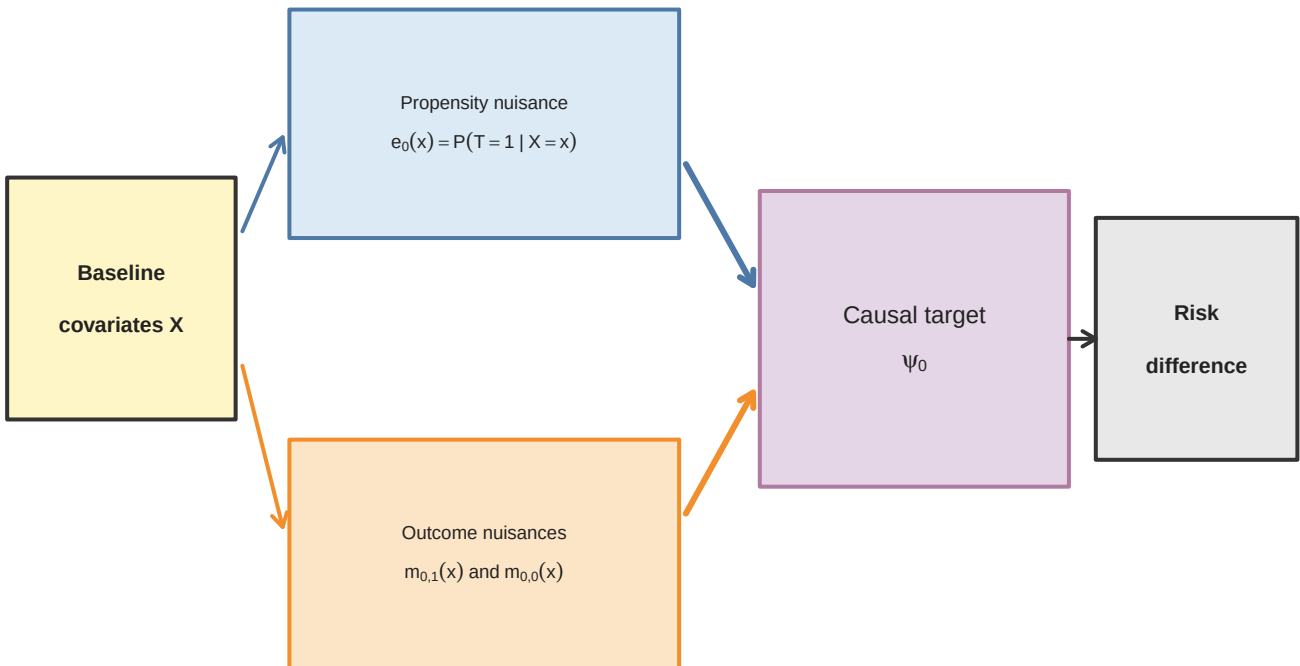
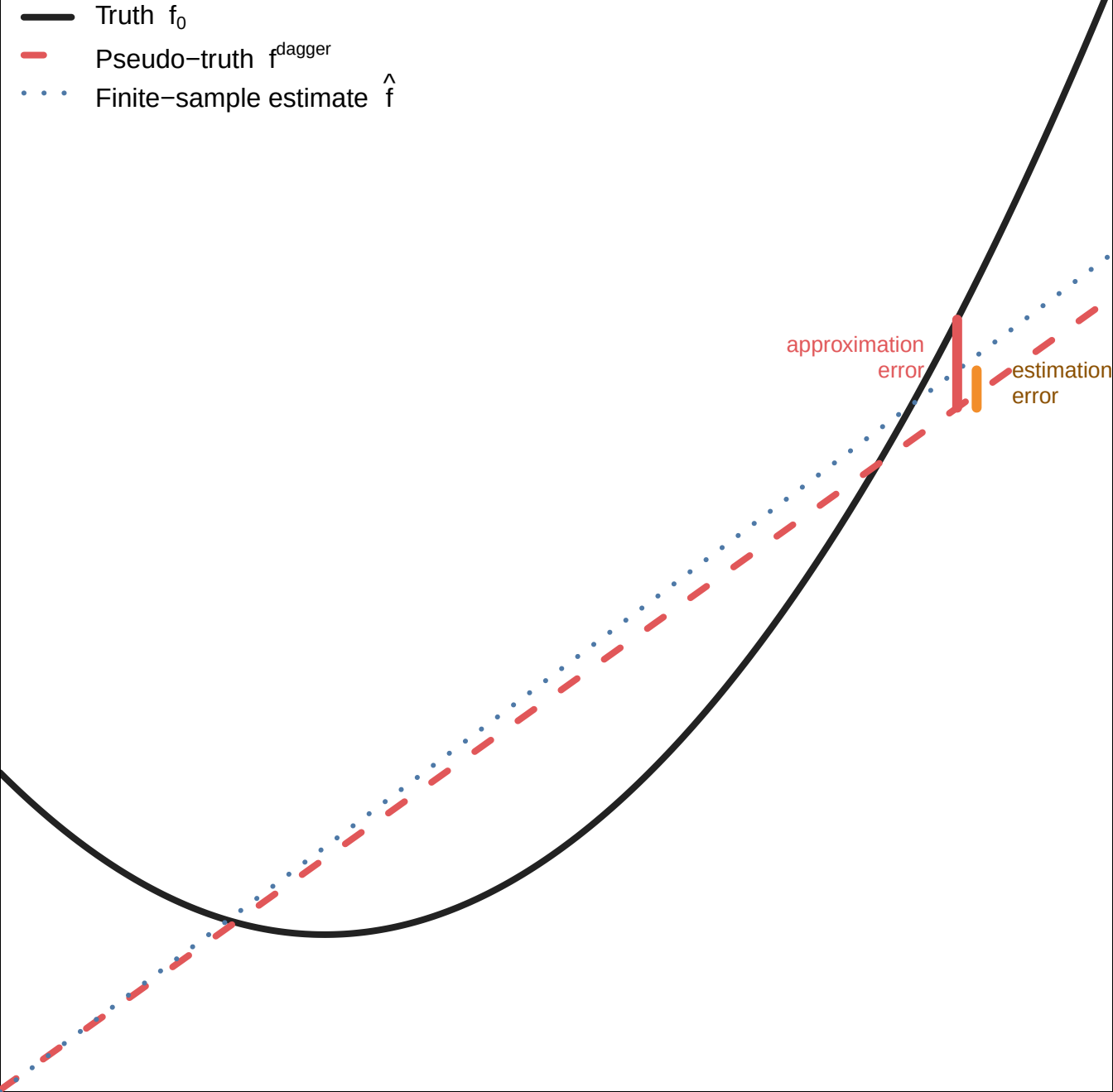




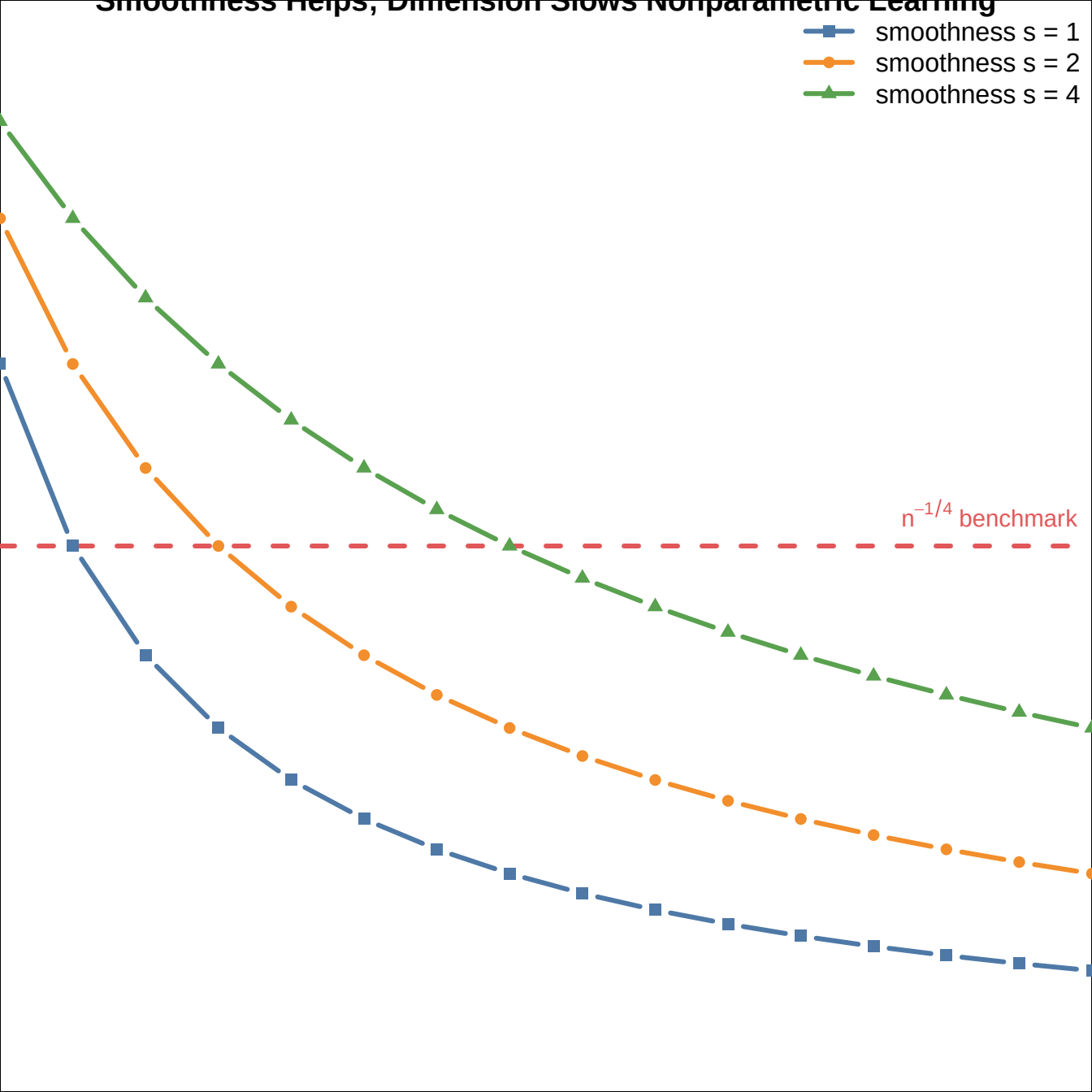


**The nuisance functions support estimation of the causal target;
they are not the final scientific estimand.**

A Fast Estimate Can Converge to the Wrong Function

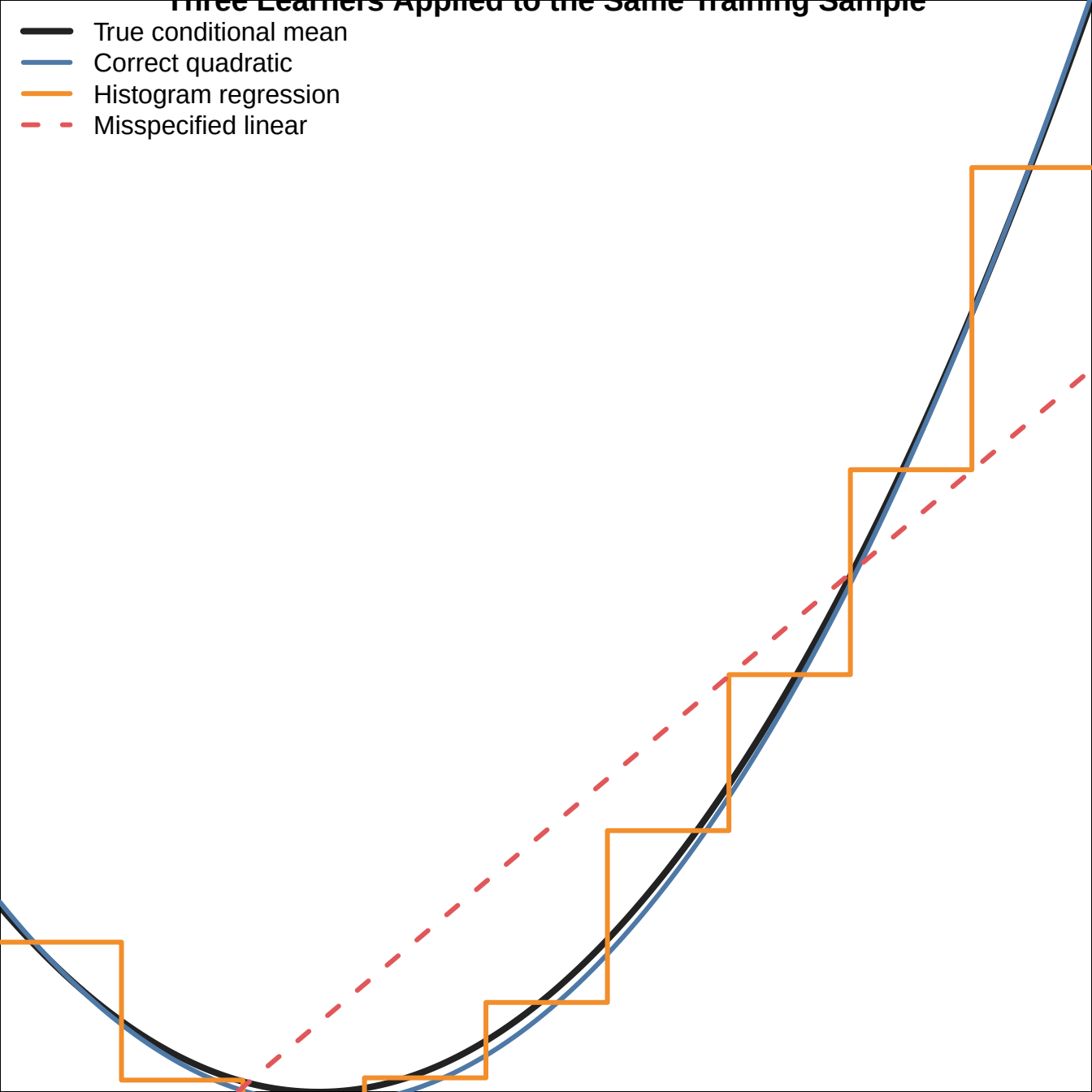


Smoothness Helps, Dimension Slows Nonparametric Learning

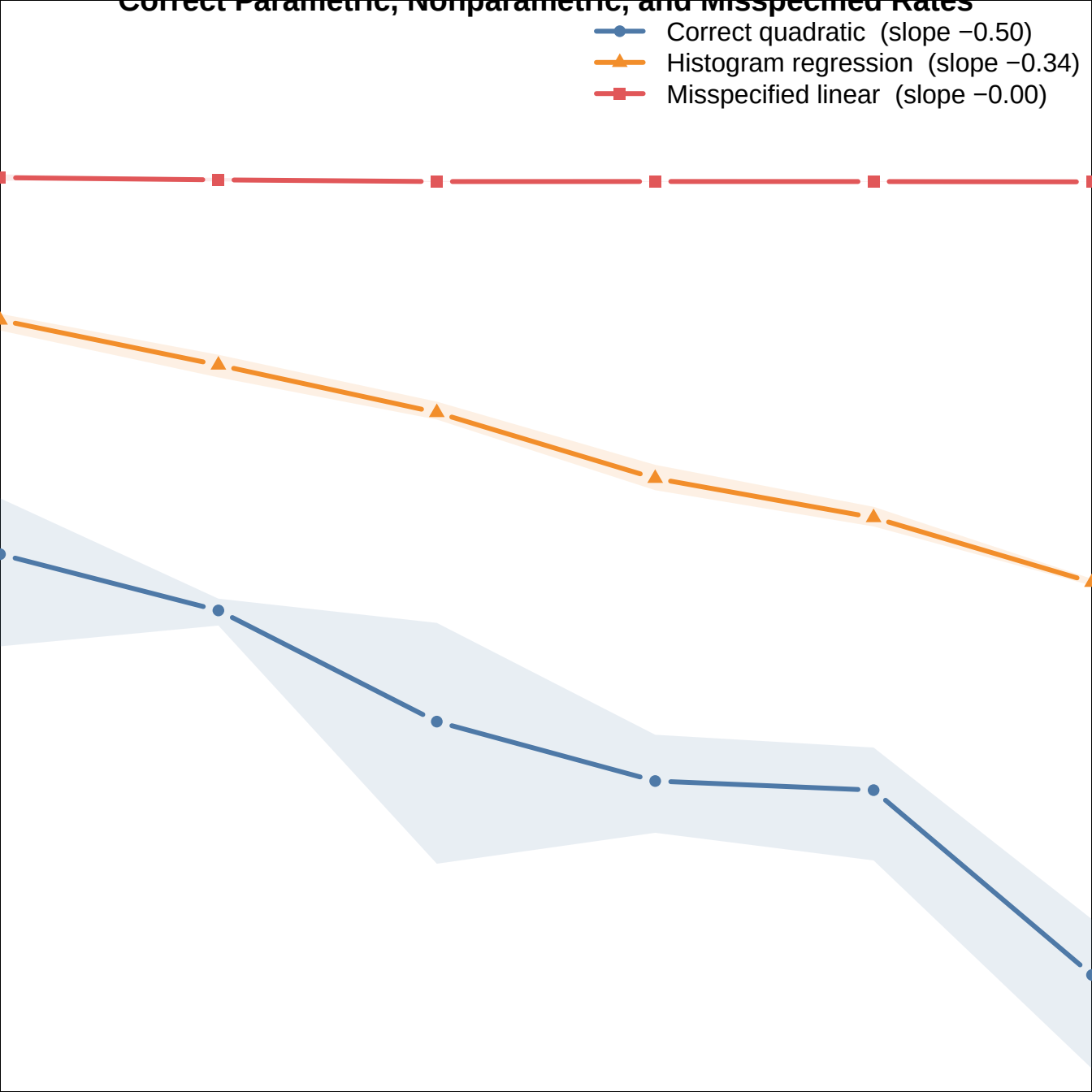


Three Learners Applied to the Same Training Sample

- True conditional mean
- Correct quadratic
- Histogram regression
- Misspecified linear



Correct Parametric, Nonparametric, and Misspecified Rates



Second-order remainder = product of nuisance errors

Propensity error

$$|\hat{e} - e_0|_2$$

X

Outcome error

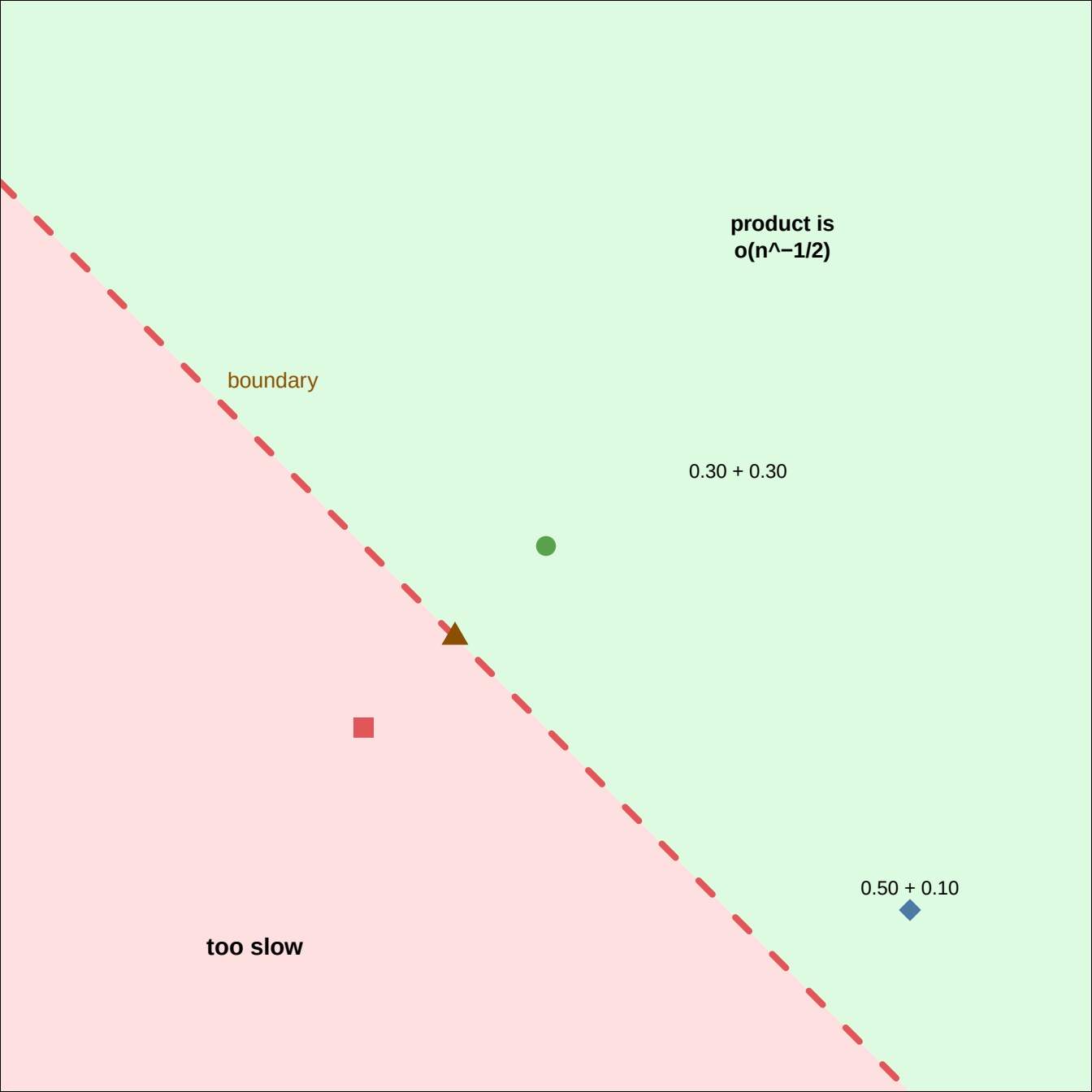
$$r_{m,n}$$

Remainder bias

$$R_2(\hat{\eta}, \eta_0)$$

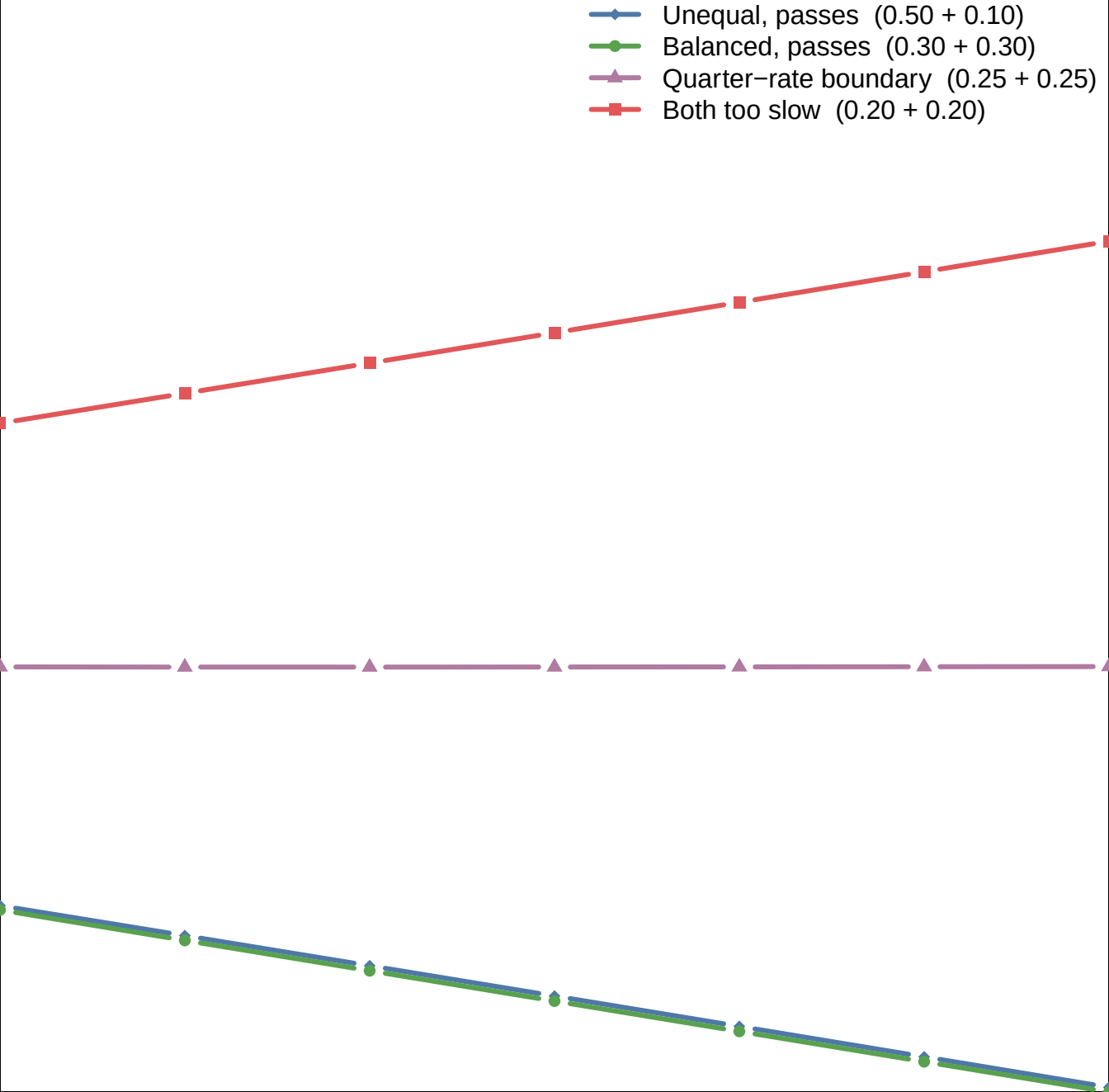
Orthogonality removes the first-order sum;

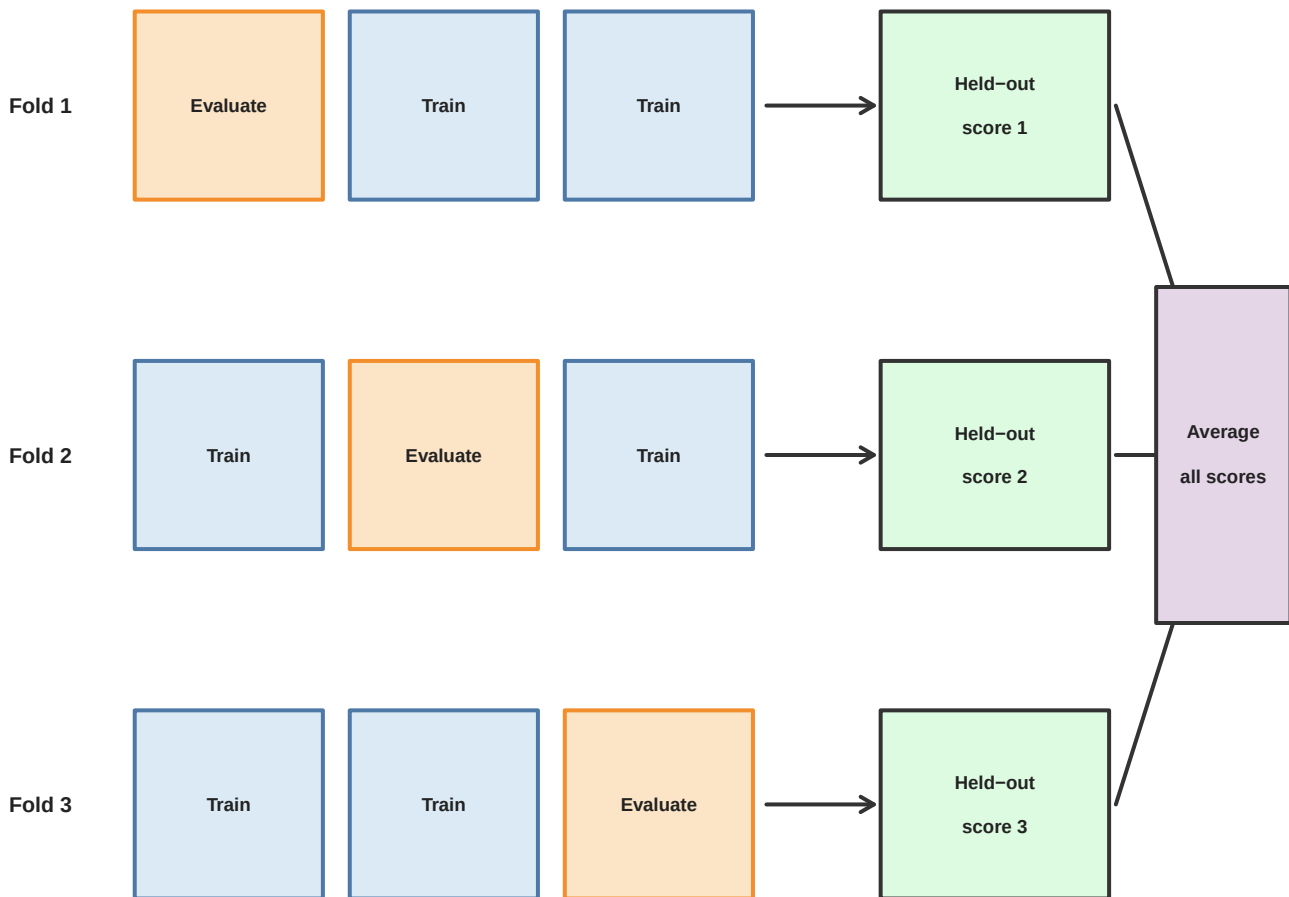
the leading drift is a product.



The Product-Rate Boundary in a Controlled Simulation

- Unequal, passes ($0.50 + 0.10$)
- Balanced, passes ($0.30 + 0.30$)
- Quarter-rate boundary ($0.25 + 0.25$)
- Both too slow ($0.20 + 0.20$)





Cross-Fitted AIPW Under Four Nuisance Specifications

True risk difference = 0.167

